

Dell Enterprise SONiC Distribution

Compatibility Matrix Release 4.4.0

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Revision history

Table 1. Revision history

Revision	Description
A00 (August 2024)	Initial release of 4.4.0

More documentation

Table 2. More documentation

Document	Description
<i>Enterprise SONiC Distribution by Dell Technologies User Guide Release</i>	Software configuration
<i>Enterprise SONiC Distribution by Dell Technologies Quick Start Guide</i>	Installation and initial configuration
<i>Enterprise SONiC Distribution by Dell Technologies Management Framework CLI Reference Guide</i>	Management Framework CLI command syntaxes and examples
<i>Enterprise SONiC Distribution by Dell Technologies Release Notes</i>	Features, requirements, known issues, and fixed issues
<i>Enterprise SONiC Distribution by Dell Technologies High Power Optics Technical Sheet</i>	Information about high-power optics

Introduction

What is SONiC

SONiC is an open-source, Linux-based network operating system (NOS) that runs on switches from multiple vendors and ASICs. SONiC stands for Software for Open Networking in the Cloud. It implements standard Layer 2 and Layer 3 protocols, and provides developers with a straightforward way to add new features.

SONiC offers teams the flexibility to create data center networking solutions, while using the collective strength of a large ecosystem with an active developer community. SONiC is designed for scalability and is in production today in large data center fabrics.

Benefits of SONiC include:

- Hardware independence
- Containerized architecture
- Open-sourced
- Access to a growing community

SONiC and Dell Technologies

Dell Technologies introduces Enterprise SONiC Distribution as a hardened, validated, and supported version of SONiC for switch configuration and monitoring. It includes distribution of open-source community SONiC, and additional features to support the ecosystem and partners.

Enterprise SONiC supports an intuitive command-line interface, and object-based administration through a REST interface and gRPC Network Management Interface (gNMI) of Google.

Enterprise SONiC Distribution by Dell Technologies

Enterprise SONiC is offered in bundles to meet customer network requirements.

- Cloud standard
- Cloud premium
- Enterprise standard
- Enterprise premium
- Lite

 **NOTE:** For supported cables and optics support matrix, reach out to Dell Technologies designated sales representative.

Table 3. Speeds supported in Enterprise SONiC on Dell PowerSwitch Platforms

Dell PowerSwitch	800G	400G	200G	100G	50G	40G	25G	10G	5G	2.5G	1G	100M	10M
Z9864F-ON	Default	Yes	Yes	Yes	No	No	No	Yes	No	No	Yes	No	No
Z9664F-ON	No	Default	Yes	Yes	No	Yes	Yes	Yes	No	No	Yes	No	No
Z9432F-ON	No	Default	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes	No	No
Z9332F-ON	No	Default	Yes	Yes	No	Yes	Yes	Yes	No	No	Yes	No	No
Z9264F-ON	No	No	No	Default	No	Yes	Yes	Yes	No	No	Yes	No	No
S5448F-ON	No	Yes	Yes	Default	Yes	Yes	Yes	Yes	No	No	Yes	No	No
S5296F-ON	No	No	No	Yes	No	Yes	Default	Yes	No	No	Yes	No	No
S5248F-ON	No	No	No	Yes	No	Yes	Default	Yes	No	No	Yes	No	No

Table 3. Speeds supported in Enterprise SONiC on Dell PowerSwitch Platforms (continued)

Dell PowerSwitch	800G	400G	200G	100G	50G	40G	25G	10G	5G	2.5G	1G	100M	10M
S5232F-ON	No	No	No	Default	Yes	Yes	Yes	Yes	No	No	Yes	No	No
S5224F-ON	No	No	No	Yes	No	Yes	Default	Yes	No	No	Yes	No	No
S5212F-ON	No	No	No	Yes	No	Yes	Default	Yes	No	No	Yes	No	No
S3248T-ON	No	No	No	Yes	No	Yes	Yes	Yes	No	No	Default	Yes	Yes
N3248P-ON*	No	No	No	Yes	No	Yes	Yes	Yes	No	No	Default	Yes	Yes
N3248PXE-ON*	No	No	No	Yes	No	Yes	Yes	Default	Yes	Yes	Yes	Yes	Yes
N3248X-ON	No	No	No	Yes	No	Yes	Yes	Default	Yes	Yes	Yes	Yes	Yes
N3248TE-ON	No	No	No	Yes	No	Yes	Yes	Yes	No	No	Default	Yes	Yes
E3248PXE-ON*	No	No	No	Yes	No	Yes	Yes	Default	Yes	Yes	Yes	Yes	Yes
E3248P-ON*	No	No	No	Yes	No	Yes	Yes	Yes	No	No	Default	Yes	Yes

NOTE:

- Platforms that are marked with the * symbol support PoE.

Dell PowerSwitches supported in each bundle

The Lite bundle is supported only on S3248T-ON, N3248P-ON, N3248TE-ON, N3248PXE-ON, N3248X-ON, E3248P-ON, and E3248PXE-ON switches. These Dell PowerSwitch platforms cannot run the Cloud and Enterprise bundles. All other Dell PowerSwitch platforms cannot run the Lite bundle.

Table 4. Dell PowerSwitches supported in each bundle

Enterprise SONiC bundle	E-Series: E3248PXE-ON, E3248P-ON	S-Series: S3248T-ON	N-Series: N3248P-ON, N3248PXE-ON, N3248TE-ON, N3248X-ON	S-Series: S5296F-ON, S5232F-ON, S5248F-ON, S5224F-ON, S5212F-ON, S5448F-ON	Z-Series: Z9264F-ON, Z9332F-ON, Z9432F-ON, Z9664F-ON, Z9864F-ON
Cloud Standard	Not Supported	Not Supported	Not Supported	Supported	Supported
Cloud Premium	Not Supported	Not Supported	Not Supported	Supported	Supported
Enterprise Standard	Not Supported	Not Supported	Not Supported	Supported	Supported
Enterprise Premium	Not Supported	Not Supported	Not Supported	Supported	Supported
Lite	Supported	Supported	Supported	Not Supported	Not Supported

Enterprise SONiC features in bundles

Table 5. Features not available in all bundles

Enterprise SONiC feature	Cloud Standard	Cloud Premium	Enterprise Standard	Enterprise Premium	Lite
ACL-based replication	Not Supported	Not Supported	Supported	Supported	Not Supported
Buffer thresholds	Supported	Supported	Supported	Supported	Not Supported
Debug application	Not Supported	Not Supported	Not Supported	Supported	Not Supported
Dot1x authentication	Not Supported	Not Supported	Not Supported	Not Supported	Supported
Dynamic ACLs	Not Supported	Not Supported	Not Supported	Not Supported	Supported
Dynamic load balancing	Supported	Supported	Supported	Supported	Not Supported
gNMI	Supported	Supported	Supported	Supported	Supported on E-Series and the S3248T-ON platform.
Guest VLANs	Not Supported	Not Supported	Not Supported	Not Supported	Supported
IGMP and IGMP snooping	Not Supported	Not Supported	Supported	Supported	Supported
Inband flow analytics (IFA)	Not Supported	Supported	Not Supported	Supported	Not Supported
IPv4 PIM source-specific multicast (SSM)	Not Supported	Not Supported	Supported	Supported	Supported
MAC authentication bypass	Not Supported	Not Supported	Not Supported	Not Supported	Supported
Multiple Spanning Tree Protocol (MSTP)	Not Supported	Not Supported	Supported	Supported	Supported
Multisite data center interconnect for VXLAN EVPN	Not Supported	Not Supported	Supported	Supported	Not Supported
Network Address Translation (NAT)	Supported	Supported	Supported	Supported	Not Supported
Packet drop monitoring	Not Supported	Not Supported	Not Supported	Supported	Not Supported
Power over Ethernet (PoE)	Not Supported	Not Supported	Not Supported	Not Supported	Supported
PVST and RPVST+	Not Supported	Not Supported	Supported	Supported	Supported
Q-in-Q VLAN tunneling	Not Supported	Not Supported	Supported	Supported	Supported
RADIUS-provided VLANs	Not Supported	Not Supported	Not Supported	Not Supported	Supported
RoCEv2	Supported	Supported	Supported	Supported	Not Supported
Tailstamping	Not Supported	Supported	Not Supported	Supported	Not Supported
Unauthenticated VLANs	Not Supported	Not Supported	Not Supported	Not Supported	Supported
VLAN translation	Not Supported	Not Supported	Supported	Supported	Supported
Voice VLANs	Not Supported	Not Supported	Not Supported	Not Supported	Supported
VXLAN	Not Supported	Not Supported	Supported	Supported	Supported
VXLAN EVPN Multihoming	Not Supported	Not Supported	Supported	Supported	Supported

Dell Technologies PowerSwitch specifications

Table 6. PowerSwitch specifications

Model	User ports	Uplinks	RAM/SSD	NPU
Z9864F-ON	64x800G QSFP112-DD	2x10G SFP+	32GB/128GB	Tomahawk5
Z9664F-ON	64x400G QSFP56-DD	2x10G SFP+	32GB/128GB	Tomahawk4
Z9432F-ON	32x400G QSFP56-DD	2x10G SFP+	32GB/64GB	Trident4.x11
Z9332F-ON	32x400G QSFP56-DD	2x10G SFP+	16GB/64GB	Tomahawk3
Z9264F-ON	64x100G QSFP28	2x10G SFP+	16GB/32GB	Tomahawk2
S5448F-ON	48x100G SFP56-DD	8x400G QSFP56-DD	32G/128GB	Trident4.x9
		2x10G SFP+		
S5232F-ON	32x100G QSFP28	2x10G SFP+	16GB/64GB	Trident3.x7
S5296F-ON	96x25G SFP28	8x100G QSFP28	16GB/64GB	Trident3.x7
S5248F-ON	48x25G SFP28	4x100G QSFP28	16GB/64GB	Trident3.x7
		2x(2x100G) QSFP28-DD		
S5224F-ON	24x25G SFP28	4x100G QSFP28	8GB/32GB	Trident3.x5
S5212F-ON	12x25G SFP28	3x100G QSFP28	8GB/16GB	Trident3.x5
S3248T-ON	48x1GBase-T	4x10G SFP+	16GB/32GB	Trident3.x3
		2x100G QSFP28		
E3248P-ON	48x1GBase-T	4x10G SFP+	16GB/32GB	Trident3.x3
		2x100G QSFP28		
E3248PXE-ON	48x10GBase-T	4x25G SFP28	16GB/32GB	Trident3.x5
		2x100G QSFP28		
N3248P-ON	48x1GBase-T	4x10G SFP+	4GB/8GB	Trident3.x3
		2x100G QSFP28		
N3248X-ON	48x10GBase-T	4x25G SFP28	4GB/8GB	Trident3.x5
		2x100G QSFP28		
N3248PXE-ON	48x10GBase-T	4x25G SFP28	4GB/8GB	Trident3.x5
		2x100G QSFP28		
N3248TE-ON	48x1GBase-T	4x10G SFP+	4GB/32GB	Trident3.x3
		2x100G QSFP28		

Standard and premium feature matrix

Table 7. Standard feature matrix

-	Features	Z9864F-ON	Z9664F-ON	Z9432F-ON	Z9332F-ON	Z9264F-ON	S5448F-ON	S5232F-ON S5248F-ON S5296F-ON	S5224F-ON S5212F-ON	N3248X-ON N3248PXE-ON E3248PXE-ON	N3248P-ON E3248P-ON N3248TE-ON S3248TE-ON
Layer 2	STP - PVST/ RPVST+	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	PVST/RPVST+ over MCLAG	No	No	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
	Link aggregation	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Link aggregation fallback	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	LACP individual	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	LLDP	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	UDLD	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	IGMP snooping	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	ACL-based Layer 2 forwarding	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	DHCP snooping	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Q-in-Q	No	No	Yes	No	No	Yes	Yes	Yes	Yes	Yes
	VLAN stacking	No	No	Yes	No	No	Yes	Yes	Yes	Yes	Yes
	Port security	No	No	No	No	No	Yes	Yes	Yes	Yes	Yes
Layer 3	BGPv4/v6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	IPv4/v6 static routing	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	BFD	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	VRRP	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	ECMP	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	DHCP relay	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	OSPFv2	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	NAT	No	No	No	No	Yes	No	Yes	Yes	No	No
	L3 IGMP	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	PIM SSM	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	IP SLA (ICMP/TCP track)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No

Table 7. Standard feature matrix (continued)

-	Features	Z9864F-ON	Z9664F-ON	Z9432F-ON	Z9332F-ON	Z9264F-ON	S5448F-ON	S5232F-ON S5248F-ON S5296F-ON	S5224F-ON S5212F-ON	N3248X-ON N3248PXE-ON E3248PXE-ON	N3248P-ON E3248P-ON N3248TE-ON S3248T-ON
	Policy-based routing	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No
	Routed sub-interfaces	No	No	Yes	No	No	Yes	Yes	Yes	Yes	No
	Adaptive routing (ARS/DLB)	Yes	Yes	No	No	No	No	No	No	No	No
	QPN hashing	Yes	Yes	Yes	No	No	No	No	No	No	No
	Flow-based hashing	Yes	Yes	No	No	No	No	No	No	No	No
	Versatile hashing	Yes	Yes	No	No	No	No	No	No	No	No
ACLs	L2 ACLs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	L3 ACLs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Receive ACLs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Route-maps	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	ACL-based replication	No	No	Yes	Yes	Yes	Yes	Yes	Yes	No	No
QoS	L2 QoS maps	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	L3 QoS maps	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Queue and buffer size configuration	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Traffic priority scheduling (strict, WFQ)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	WRED	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	ECN	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Priority flow control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	BUM/storm control	No	No	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
	RoCE V2	Yes	Yes	Yes	Yes	No	No	Yes	No	No	No
Mgmt and monitoring	Telnet	No	No	No	No	No	No	No	No	No	No
	SSH	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	SCP	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	TFTP	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	FTP	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	NTP	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	SNMPv2/v3	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	ZTP	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 7. Standard feature matrix (continued)

-	Features	Z9864F-ON	Z9664F-ON	Z9432F-ON	Z9332F-ON	Z9264F-ON	S5448F-ON	S5232F-ON S5248F-ON S5296F-ON	S5224F-ON S5212F-ON	N3248X-ON N3248PXE-ON E3248PXE-ON	N3248P-ON E3248P-ON N3248TE-ON S3248T-ON
	sFlow	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
MCLAG	L2	No	No	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
	L3	No	No	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
	L2 Logical VTEP	No	No	Yes	No	No	Yes	Yes	Yes	Yes	Yes
	L3 Logical VTEP	No	No	Yes	No	No	Yes	Yes	Yes	Yes	Yes
	IGMP snooping	No	No	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
	MCLAG graceful shutdown	No	No	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
	MCLAG fallback	No	No	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
	MCLAG LACP individual	No	No	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
EVPN	L2 VXLAN	No	No	Yes	No	No	Yes	Yes	Yes	Yes	Yes
	L3 VXLAN	Yes	No	Yes	No	No	Yes	Yes	Yes	Yes	Yes
	Data Center Interconnect (DCI)	No	No	Yes	No	No	Yes	Yes	Yes	No	No
	Multihoming	No	No	Yes	No	No	Yes	Yes	Yes	Yes	Yes
Security	TACACS+	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	RADIUS	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	RBAC	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
System	Dynamic port breakout	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Interface aliasing	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Display running configuration using CLI	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	User container management	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No
	Warm restart	No	No	No	Yes	Yes	No	Yes	Yes	No	No
Telemetry	Thresholds	No	No	No	Yes	Yes	No	Yes	Yes	No	No
	Snapshots	No	No	No	Yes	Yes	No	Yes	Yes	No	No

Table 8. Premium feature matrix

-	Features	Z9864F-ON	Z9664F-ON	Z9432F-ON	Z9332F-ON	Z9264F-ON	S5448F-ON	S5232F-ON S5248F-ON S5296F-ON	S5224F-ON S5212F-ON	N3248P-ON, N3248X-ON, N3248PXE-ON, E3248PXE-ON, S3248T-ON	N3248TE-ON
Silicon telemetry	Drop monitor	No	No	No	Yes	No	No	Yes	No	No	No
	Inband flow analyzer (IFA v2.0)	No	No	No	No	No	No	Yes	No	No	No
	Tailstamping	No	No	No	Yes	Yes	No	Yes	No	No	No

Scalability

Table 9. Z-series scalability

-	Z9864F-ON	Z9664F-ON	Z9432F-ON	Z9332F-ON	Z9264F-ON
Number of port channels per system	128	128	128	128	128
Number of port channel members per system	128	128	128	128	128
Maximum number of port channel members per port channel	32	32	32	32	32
Maximum number of PVST instances	62	62	254	62	510
Maximum number of RPVST instances	62	62	254	62	510
Maximum number of RPVST VLAN ports	5000	5000	5000	5000	5000
Number of VLANs per system	3965	3965	3965	3965	3965
Maximum number of VLAN members per VLAN	Depends on the maximum number of switch ports.				
Number of VLAN members per system	23790	23790	15K(with VLAN stacking)	23790	23790
			23790(without VLAN stacking)		
Maximum number of SAG per system	512	512	3965	512	1k
Maximum number of MLAG unique IP VLAN L3 interfaces (SVI) per system	Not supported	Not supported	3965	Not supported	1k
Maximum number of VLAN L3 interfaces (SVI) per system (if VRRP, SAG, and MLAG unique IP interfaces are not configured)	3965	3965	3965	3965	3965
Maximum number of Layer 3 interfaces on physical, port channel, and subinterface ports supported per system	512	512	1k	512	1k
Maximum number of MAC addresses per system	8k/8k *	8k	32k	8k/8k *	40k/256k *
Maximum number of VRFs per system	1000	1000	1000	512	1000
Maximum number of IPv4 routes per system (default mode)	1M	1M	1M	196k	196k
Maximum number of IPv6 routes per system (default mode by prefix length)	1M	1M	1M	32k (<=64b)	32k (<=64b)
				25k (>64b)	25k (>64b)

Table 9. Z-series scalability (continued)

-	Z9864F-ON	Z9664F-ON	Z9432F-ON	Z9332F-ON	Z9264F-ON
Maximum number of IPv4 routes per system (route scale maximum mode)	Not supported	Not supported	Not supported	Not supported	280k
Maximum number of IPv6 routes per system (route scale maximum mode)	Not supported	Not supported	Not supported	Not supported	32k (<=64b, >64b)
Maximum number of static routes per system	2k	2k	2k	2k	2k
Maximum number of BGP routes per system	1M	1M	1M	IPv4 - 196k	IPv4 default mode - 196k
					IPv6 default mode - 32k (<=64b), 25k (>64b)
				IPv6 - 32k (<=64b), 25k (>64b)	IPv4 route scale mode - 280k
					IPv6 route scale mode - 32k (<=64b, >64b)
Maximum number of BGP routes RIB IN	1M	1M	1M	IPv4 - 196k	IPv4 default mode - 196k
					IPv6 default mode - 32k (<=64b), 25k (>64b)
					IPv4 route scale mode - 280k
				IPv6 - 32k (<=64b), 25k (>64b)	IPv6 route scale mode - 32k (<=64b, >64b)
Maximum number of BGP neighbors	512	512	512	512	512
Maximum number of ECMP groups	1023	1023	1023	1023	1023
Maximum number of paths in an ECMP group	128	128	128	128	128
Maximum number of BFD sessions	128	128	128	128	128
IP SLA - ICMP tracker	50	50	50	50	50
IP SLA - TCP tracker	50	50	50	50	50
Maximum number of ARP supported per system	16k	16k	30k	16k	32k
Maximum number of ND supported per system	8k	8k	15k	8k	16k
Maximum number of OSPF routers per OSPF area	50	50	50	50	50
Maximum number of OSPF-enabled L3 interfaces	128	128	128	128	128

Table 9. Z-series scalability (continued)

-	Z9864F-ON	Z9664F-ON	Z9432F-ON	Z9332F-ON	Z9264F-ON
Maximum number of OSPFv2 neighbors	128	128	128	128	128
Maximum number of OSPF intra-area routes	5,000	5,000	5,000	5,000	5,000
Maximum number interarea summary routes	5,000	5,000	5,000	5,000	5,000
Maximum number of OSPF external (type-5) routes	40,000	40,000	40,000	40,000	40,000
Maximum number of VRRP instances	128	128	128	128	128
Maximum number of VRRP enabled interfaces	128	128	128	128	128
Maximum number of VRRP instances per interface	16	16	16	16	16
Maximum number of tracked interfaces per VRRP instance	8	8	8	8	8
Maximum IP address per VRRP instance	4	4	4	4	4
Maximum number of IPv4 only ACL tables that are applied to interfaces at ingress	2	2	2	2	3
Maximum number of IPv4 ACL rules per applied ACL table at ingress	255	255	1023	256**	256
Maximum number of IPv6 only ACL tables that are applied to interfaces at ingress	2	2	2	1	1
Number of IPv6 ACL rules per applied ACL table at ingress	255	255	1023	256**	256
Maximum number of IPv4 only ACL tables applied to interfaces at egress	1	1	1	1	1
Maximum number of IPv4 ACL rules per applied ACL tables at egress	127	127	511	128	256
Maximum number of IPv6 only ACL tables applied to interfaces at egress	1	1	1	1	1
Number of IPv6 ACL rules per applied ACL tables at egress	127	127	511	128	256
Maximum number of egress MAC only ACL tables/features applied to an interface	1	1	1	1	1
Maximum number of MAC ACL rules per applied ACL table/feature at egress	127	127	511	128	256
Maximum number of ingress MAC only ACL tables/features applied to an interface	2	2	2	2	3

Table 9. Z-series scalability (continued)

-	Z9864F-ON	Z9664F-ON	Z9432F-ON	Z9332F-ON	Z9264F-ON
Maximum number of MAC ACL rules per applied ACL table/feature at Ingress	255	255	1023	256	256
Maximum number of MST instances	61	61	64	61	61
Maximum number of VLAN ports for MST	5000	5000	5000	5000	5000
Maximum number of NAT entries (static plus dynamic)	0	0	0	0	1024
Maximum number of port mirroring sessions per system	3	3	3	3	3
Maximum number of SPAN/ERSPAN mirroring sessions per system	3	3	3	3	3
Maximum number of BGP EVPN sessions	256	256	256	256	256
Maximum EVPN MAC addresses	Not supported	Not supported	32k	Not supported	Not supported
Maximum VXLAN tunnels	512 (Without primary IP addresses (PIP))	Not supported	512 (Without primary IP addresses (PIP)) 256 (With PIP)	Not supported	Not supported
	256 (With PIP)				
Maximum number of L2 VNIs	Not supported	Not supported	3965	Not supported	Not supported
Maximum number of L3 VNIs (VRF)	1k	Not supported	1k	Not supported	Not supported
Maximum number of IPMC (L3) forwarding entries	512	512	8k	512	8k
Overlay routes	IPv4 - 1M	Not supported	IPv4 - 340k	Not supported	Not supported
	IPv6 - 1M		IPv6 - 340k		
Maximum number of flow groups for silicon telemetry features	Not supported	Not supported	Not supported	253	253
Maximum number of IFA sessions	Not supported	Not supported	Not supported	Not supported	Not supported
Maximum number of drop monitor sessions	Not supported	Not supported	Not supported	253	Not supported
Maximum number of tailstamping sessions	Not supported	Not supported	Not supported	Not supported	253
Maximum number of IGMP Snooping entries	512	512	512	512	512
Max number of DHCPv4 Snooping entries (dynamic/static)	12k / 1k	12k / 1k	12k / 1k	12k / 1k	12k / 1k
Max number of DHCPv6 Snooping entries (dynamic/static)	12k / 1k	12k / 1k	12k / 1k	12k / 1k	12k / 1k
PAC : Maximum Clients (802.1x/MAB) supported on the Switch	Not supported	Not supported	Not supported	Not supported	Not supported
PAC : Maximum Clients (802.1x/MAB) supported per port	Not supported	Not supported	Not supported	Not supported	Not supported

Table 9. Z-series scalability (continued)

-	Z9864F-ON	Z9664F-ON	Z9432F-ON	Z9332F-ON	Z9264F-ON
PAC : Maximum ACL rules supported per (802.1x/MAB) Client	Not supported	Not supported	Not supported	Not supported	Not supported
Q-in-Q single tag	Not supported	Not supported	1983	Not supported	Not supported
Q-in-Q double tag	Not supported	Not supported	1983	Not supported	Not supported
Maximum number of active PBR entries per system	510	510	2046	512	2304
EVPN multihoming maximum ESIs per VTEP	Not supported	Not supported	128	Not supported	Not supported
EVPN multihoming maximum VNIs per ESI	Not supported	Not supported	4k	Not supported	Not supported
EVPN multihoming maximum VTEPs in an ES	Not supported	Not supported	4	Not supported	Not supported
Routed subinterfaces	Not supported	Not supported	750	Not supported	Not supported

Table 10. E, N, S-series scalability

-	S5448F-ON	S5232F-ON, S5248F-ON, S5296F-ON	S5224F-ON, S5212F-ON	N3248P-ON, E3248P-ON, N3248TE-ON, S3248T-ON	N3248PXE-ON, E3248PXE-ON, N3248X-ON
Number of port channels per system	128	128	128	8	8
Number of port channel members per system	128	128	128	64	64
Maximum number of port channel members per port channel	32	32	32	8	8
Maximum number of PVST instances	254	510	510	510	510
Maximum number of RPVST instances	254	510	510	510	510
Maximum number of RPVST VLAN ports	5000	5000	5000	1000	1000
Number of VLANs per system	3965	3965	3965	3965	3965
Maximum number of VLAN members per VLAN	Depends on the maximum number of switch ports.				
Number of VLAN members per system	15k (with VLAN stacking)	12k	12k	3965	3965
	23790 (without VLAN stacking)				
Maximum number of SAG per system	3965	3965	3965	1k	1k
Maximum number of MLAG unique IP VLAN L3 interfaces (SVI) per system	3965	3965	3965	1k	1k

Table 10. E, N, S-series scalability (continued)

-	S5448F-ON	S5232F-ON, S5248F-ON, S5296F-ON	S5224F-ON, S5212F-ON	N3248P-ON, E3248P-ON, N3248TE-ON, S3248T-ON	N3248PXE-ON, E3248PXE-ON, N3248X-ON
Maximum number of VLAN L3 interfaces (SVI) per system (if VRRP, SAG, and MCLAG unique IP interfaces are not configured)	3965	3965	3965	1k	1k
Maximum number of Layer 3 interfaces on physical, port channel, and subinterface ports supported per system	1k	1k	1k	1k	1k
Maximum number of MAC addresses per system	32K/224k *	64k/224k *	64k/224k *	8k	8k
Maximum number of VRFs per system	1000	1000	1000	32	32
Maximum number of IPv4 routes per system (default mode)	1M	81k	81k	8158	8158
Maximum number of IPv6 routes per system (default mode by prefix length)	1M	32k (<=64b)	32k (<=64b)	4096	4096
		25k (>64b)	25k (>64b)		
Maximum number of IPv4 routes per system (route scale maximum mode)	Not supported	160k	160k	Not supported	Not supported
Maximum number of IPv6 routes per system (route scale maximum mode)	Not supported	65k (<=64b, >64b)	65k (<=64b, >64b)	Not supported	Not supported
Maximum number of static routes per system	2k	2k	1024	1024	1024
Maximum number of BGP routes per system	1M	IPv4 default mode - 81k	IPv4 default mode - 81k	IPv4 - 8158	IPv4 - 8158
		IPv6 default mode - 32k (<=64b), 25k (>64b)	IPv6 default mode - 32k (<=64b), 25k (>64b)		
		IPv4 route scale mode - 160k	IPv4 route scale mode - 160k		
		IPv6 route scale mode - 65k (<=64b, >64b)	IPv6 route scale mode - 65k (<=64b, >64b)	IPv6 - 4096	IPv6 - 4096
Maximum number of BGP routes RIB IN	1M	IPv4 default mode - 81k	IPv4 default mode - 81k	IPv4 - 8158	IPv4 - 8158
		IPv6 default mode - 32k (<=64b), 25k (>64b)	IPv6 default mode - 32k (<=64b), 25k (>64b)		
		IPv4 route scale mode - 160k	IPv4 route scale mode - 160k		
		IPv6 route scale mode - 65k (<=64b, >64b)	IPv6 route scale mode - 65k (<=64b, >64b)	IPv6 - 4096	IPv6 - 4096

Table 10. E, N, S-series scalability (continued)

-	S5448F-ON	S5232F-ON, S5248F-ON, S5296F-ON	S5224F-ON, S5212F-ON	N3248P-ON, E3248P-ON, N3248TE-ON, S3248T-ON	N3248PXE-ON, E3248PXE-ON, N3248X-ON
Maximum number of BGP neighbors	512	512	256	128	128
Maximum number of ECMP groups	1023	1023	1023	256	256
Maximum number of paths in an ECMP group	128	128	128	16	16
Maximum number of BFD sessions	128	128	64	64	64
IP SLA - ICMP tracker	50	50	50	25	50
IP SLA - TCP tracker	50	50	50	25	50
Maximum number of ARP supported per system	30k	32k	16k	6144	6144
Maximum number of ND supported per system	15k	16k	8k	2560	2560
Maximum number of OSPF routers per OSPF area	50	50	50	50	50
Maximum number of OSPF-enabled L3 interfaces	128	128	128	64	64
Maximum number of OSPFv2 neighbors	128	128	128	64	64
Maximum number of OSPF intra-area routes	5,000	5,000	5,000	2,500	2,500
Maximum number interarea summary routes	5,000	5,000	5,000	2,500	2,500
Maximum number of OSPF external (type-5) routes	40,000	40,000	20,000	8k	8k
Maximum number of VRRP instances	128	128	128	64	64
Maximum number of VRRP enabled interfaces	128	128	128	64	64
Maximum number of VRRP instances per interface	16	16	16	8	8
Maximum number of tracked interfaces per VRRP instance	8	8	8	8	8
Maximum IP address per VRRP instance	4	4	4	4	4
Maximum number of IPv4 only ACL tables that are applied to interfaces at ingress	2	3	3	4	3
Maximum number of IPv4 ACL rules per applied ACL table at ingress	1023	768	768	512	768
Maximum number of IPv6 only ACL tables that are applied to interfaces at ingress	2	3	3	4	3

Table 10. E, N, S-series scalability (continued)

-	S5448F-ON	S5232F-ON, S5248F-ON, S5296F-ON	S5224F-ON, S5212F-ON	N3248P-ON, E3248P-ON, N3248TE-ON, S3248T-ON	N3248PXE-ON, E3248PXE-ON, N3248X-ON
Number of IPv6 ACL rules per applied ACL table at ingress	1023	768	768	512	768
Maximum number of IPv4 only ACL tables applied to interfaces at egress	1	1	1	1	1
Maximum number of IPv4 ACL rules per applied ACL tables at egress	511	512	512	512	512
Maximum number of IPv6 only ACL tables applied to interfaces at egress	1	1	1	1	1
Number of IPv6 ACL rules per applied ACL tables at egress	511	512	512	256	512
Maximum number of egress MAC only ACL tables/features applied to an interface	1	1	1	1	1
Maximum number of MAC ACL rules per applied ACL table/feature at egress	511	512	512	512	512
Maximum number of ingress MAC only ACL tables/features applied to an interface	2	3	3	4	3
Maximum number of MAC ACL rules per applied ACL table/feature at Ingress	1023	768	768	512	768
Maximum number of MST instances	64	64	64	64	64
Maximum number of VLAN ports for MST	5000	5000	5000	1000	1000
Maximum number of NAT entries (static plus dynamic)	0	1023	1023	Not supported	Not supported
Maximum number of port mirroring sessions per system	3	3	3	3	3
Maximum number of SPAN/ERSPAN mirroring sessions per system	3	3	3	3	3
Maximum number of BGP EVPN sessions	256	256	128	64	64
Maximum EVPN MAC addresses	32k/224k*	64k/224k*	64k/224k*	8k	8k
Maximum VXLAN tunnels	512 (Without primary IP addresses (PIP))	512 (Without primary IP addresses (PIP))	512 (Without primary IP addresses (PIP))	128	128
	256 (With PIP)	256 (With PIP)	256 (With PIP)		
Maximum number of L2 VNIs	3965	3965	3965	2k	2k

Table 10. E, N, S-series scalability (continued)

-	S5448F-ON	S5232F-ON, S5248F-ON, S5296F-ON	S5224F-ON, S5212F-ON	N3248P-ON, E3248P-ON, N3248TE-ON, S3248T-ON	N3248PXE-ON, E3248PXE-ON, N3248X-ON
Maximum number of L3 VNIs (VRF)	1k	1k	1k	32	32
Overlay routes	IPv4 - 340k	IPv4 - 81k	IPv4 - 81k	IPv4 - 8k	IPv4 - 8k
	IPv6 - 340k	IPv6 - 25k	IPv6 - 25k	IPv6 - 4K	IPv6 - 4K
Maximum number of IPMC (L3) forwarding entries	8k	8k	8k	8k	8k
Maximum number of flow groups for silicon telemetry features	Not supported	253	Not supported	Not supported	Not supported
Maximum number of IFA sessions	Not supported	249	Not supported	Not supported	Not supported
Maximum number of drop monitor sessions	Not supported	253	Not supported	Not supported	Not supported
Maximum number of tailstamping sessions	Not supported	253	Not supported	Not supported	Not supported
Maximum number of IGMP Snooping entries	512	512	512	512	512
Max number of DHCPv4 Snooping entries (dynamic/static)	12k / 1k	12k / 1k	12k / 1k	3k / 1k	3k / 1k
Max number of DHCPv6 Snooping entries (dynamic/static)	12k / 1k	12k / 1k	12k / 1k	3k / 1k	3k / 1k
PAC : Maximum Clients (802.1x/MAB) supported on the Switch	Not supported	Not supported	Not supported	512	512
PAC : Maximum Clients (802.1x/MAB) supported per port	Not supported	Not supported	Not supported	48	48
PAC : Maximum ACL rules supported per (802.1x/MAB) Client	Not supported	Not supported	Not supported	60	60
Q-in-Q single tag	1983	1983	1983	1983	1983
Q-in-Q double tag	1983	1983	1983	1983	1983
Maximum number of active PBR entries per system	2046	2304	2304	2048	2304
EVPN multihoming maximum ESIs per VTEP	128	128	128	64	64
EVPN multihoming maximum VNIs per ESI	4k	4k	4k	1k	1k
EVPN multihoming maximum VTEPs in an ES	4	4	4	4	4
Routed subinterfaces	750	750	750	Not supported	Not supported

* The first value corresponds to Layer 3 switch profile mode (default mode), and the second value corresponds to Layer 2 switch profile mode.

** For the Z9332F-ON, one ACL table can have 512 entries while the other can have 256. It is not deterministic which table is 512 when tables are added and deleted (marked as 256).

Breakout modes on Dell PowerSwitch platforms

This information lists the breakout modes for each supported switch.

Table 11. Breakout modes on Dell PowerSwitch platforms

Platform	Breakout modes	Default breakout mode
S3248T-ON	1x100G, 1x40G, 4x25G, 4x10G, 1x25G, 1x10G	The listed breakout modes are supported on ports 1/53 and 1/54.
S5212F-ON	1x100G, 1x40G, 4x25G, 4x10G	- The listed breakout modes are supported on port 1/13 - 1/15. - Port 1/1 to 1/12 do not support breakout and are 25G by default. - You can use them at 10G or 1G using the speed command.
S5224F-ON	1x100G, 1x40G, 4x25G, 4x10G	- The listed breakout modes are supported on port 1/25 - 1/28. - Port 1/1 to 1/24 do not support breakout and are 25G by default. - You can use them at 10G or 1G using the speed command.
S5232F-ON	1x100G, 1x40G, 2x50, 4x25G, 4x10G	- Listed Breakout modes are supported on port 1/1 - 1/31. - Port 1/32 does not support 4x25G or 4x10G modes. To configure Port1/32 to operate at 40G, use the speed 40000 command. 2x50G is validated only for a specific case with 100G DAC.
S5248F-ON	1x100G, 1x40G, 4x25G, 4x10G	- The listed breakout modes are supported on port 1/49 - 1/56. - Port 1/1 to 1/48 do not support breakout and are 25G by default. You can use them at 10G or 1G using the port-group command. For more information about using the port-group command, see the User Guide.
S5296F-ON	1x100G, 1x40G, 4x25G, 4x10G	- The listed breakout modes are supported on port 1/97 - 1/104. - Port 1/1 to 1/96 do not support breakout and are 25G by default. - You can use them at 10G or 1G using the port-group command. For more information about using the port-group command, see the User Guide.
S5448F-ON	Port 1/1 to 1/4 and 1/53 to 1/56: 1x25G, 1x10G, 1x100G, 1x40G,	-

Table 11. Breakout modes on Dell PowerSwitch platforms (continued)

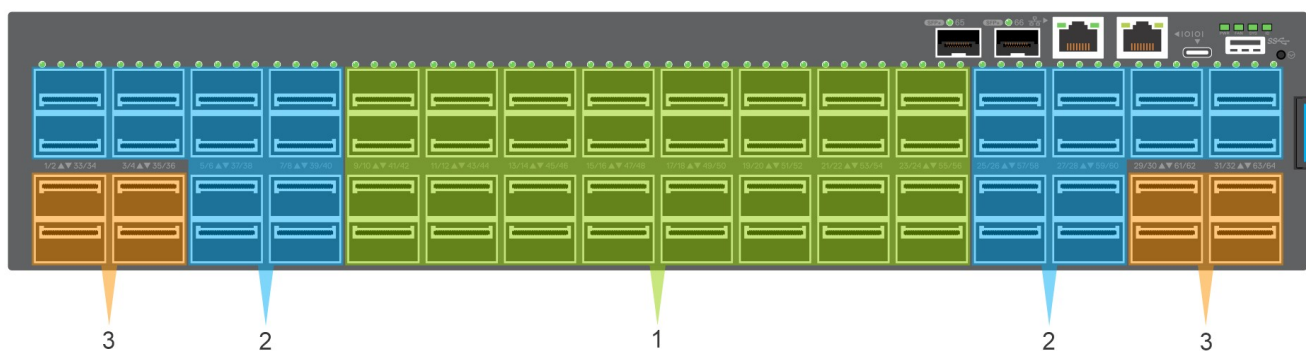
Platform	Breakout modes	Default breakout mode
	1x200G, 2x40G, 2x200G, 2x100G, 4x25G, 4x10G, 4x100G, 1x400G Port 1/5 to 1/52: 1x25G, 1x10G, 1x100G, 1x40G, 1x200G, 1x400G, 2x40G, 2x200G, 2x100G, 4x25G, 4x10G, 2x50G, 4x100G	
Z9264F-ON	1x100G, 1x40G, 4x25G, 4x10G, 1x25G, 1x10G	-
Z9332F-ON	1x400G, 2x200G, 2x100G, 2x40G, 4x100G, 1x100G, 1x40G, 1x200G, 8x25G, 8x10G, 4x25G, 4x10G, 1x25G, 1x10G	-
Z9432F-ON	1x400G, 2x200G, 2x100G, 2x40G, 4x100G, 1x100G, 1x40G, 1x200G, 8x50G, 8x25G, 8x10G, 4x25G, 4x10G, 1x25G, 1x10G	-
Z9664F-ON	1x400G, 2x200G, 2x100G, 2x40G, 4x100G, 1x100G, 1x40G, 4x25G, 4x10G, 1x25G, 1x10G	-
Z9864F-ON	1x800G, 4x200G, 2x400G, 2x200G (50G PAM4), 2x100G (25G NRZ), 8x100G	-
N and E-series	1x100G, 1x40G, 4x25G, 4x10G, 1x25G, 1x10G	The listed breakout modes are supported on ports 1/53 and 1/54.

Z9864F-ON ports and DAC cable length compatibility

The 800G OSFP112 passive copper cables come in various lengths. On the Z9864F-ON platform, certain ports have a limitation on the maximum cable length that can be used on them.

Z9864F-ON port-to-port connection

The following port map displays the groups of ports and the cable lengths that they support when connection is made to another Z9864F-ON 800G OSFP112 port. If the length is exceeded, the link may not come up or become unreliable.



When you design your system wiring to use passive copper DAC cables, ensure that you use the correct cable on each port as follows:

1. Port group 1 includes ports 9 to 24 and 41 to 56, and supports Dell-branded DAC cables 1m, 2m, 3m, and 4m.
2. Port group 2 includes ports 1 to 8, 25 to 32, 37 to 40, and 57 to 60, and supports Dell-branded DAC cables 1m, 2m, and 3m.
3. Port group 3 includes ports 33 to 36 and 61 to 64 and supports Dell-branded DAC cables 1m and 2m.

Z9864F-ON port-to-Dell Broadcom Ethernet NIC connection

If you connect the Z9864F-ON platform from any of the 1 to 64 port to a Dell Broadcom BCM57608 2x200G PCIe Ethernet NIC, ensure that you use Dell-branded passive copper DAC cables of 1m, 2m, 3m, or 4m. If the length exceeds, the link may not come up or become unreliable.